

Documentation

Leibinger-Jet3 Interface protocol

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General

The interface allows remote control and remote diagnosis access via Ethernet, RS232 and modem. The interface is based on a dual-master protocol which is able to send from both sides independently.

The protocol generates a frame around the LJscript-language so that the interface can directly transfer script-sequences.

To offer a terminal-function the protocol works without a direct package reply. For the usage of a serial interface it is required to activate the handshake-controlled hardware-flow control in the transmission range to avoid a receive-sited overflow.

All packages are started with a ^-character and are finished with the character <CR> (Hex 0D). Optionally you can add a <LF> (Hex 0A).

After the start character the destination address follows, which consists of a character in the range of 0-9 and A-F. To address a LJ3-Printer you have to use the address "0". Further addresses are not used at the moment, they are just used as reserve for later options. Afterwards a character continues which selects a command group.

These are:

Character	Command-group
*	Script-instruction (a line continues as described in LJ-script)
!	Action-instruction (Printer control, e.g. nozzle open/closed, Print start, test print etc.)
?	Inquiry (The designation of the instruction which has been inquired continues e.g. ?HS inquires the current hydraulic-settings)
=	Parameter Transfer (e.g. =HS7100 2400 is the response of the machine to ?HS inquiry)
\$	File Transfer, Directory Infos

The transmitted data within the frame are ASCII-data which should not include the codes ^ (0x5e) and <CR> (0x0d). If the character ^ should occur in the data it has to be designated with a prepend backslash „\“ (\ ^).

Variables are passed as ASCII-coded decimal values which are separated by TABs if nothing else has been specified.

The following definitions of the commands are only valid for the turned on device. If the device is in standby mode, only the instruction !PO (Power ON), !RM (Reset) or FileTransfer can be handled.

Script Processing

The instructions which are described in the LJ-script can be transferred directly with Frame-Header.

A script starts with

`^0*BEGINLJSCRIPT<CR>`

and ends with

`^0*ENDLJSCRIPT<CR>`

As soon as the instruction BEGINLJSCRIPT has been transmitted, the current job data file (that means the work file which is loaded during the reboot of the system) is deleted and will be overwritten with the following script instructions. As soon as the ENDLJSCRIPT instruction has been transmitted the data file is passed to the RIP for printing. In this situation ****EXTERN**** is displayed as job name.

<CR> is the character Carriage Return and has the value Hex 0D (0x0D).

You will find details regarding the LJ3 script language in the „Documentation LJ3 Script language“.

Action instructions

Action instructions are commands which are transferred without parameters and which carry out a direct reaction in the printer. The following actions are implemented.

Overview

Command	Description
!GO	Start Print-Ready
!ST	Stop Print-Ready
!NC	Nozzle Close
!NO	Nozzle Open
!RC	Reset Counter (Object Counter)
!FA	Factory Reset
!MC	Motor Step Close
!MO	Motor Step Open
!LF	Re-Load Fonts
!OK	CRC OK (Host←Printer)
!US	User Suspend (Lock of printer operation)
!UR	User Resume (Release of printer operation)
!EQ	Error Quit
!PO	Power ON (Out of Standby)
!PF	Power OFF (in Standby mode)
!EN	End of Data (Host←Printer)
!RM	Restart Machine (= Software Reset, only while Standby)
!W0	Wysiwyg display off
!W1	Wysiwyg display on
!FF	Flush Mail Fifo (Clearing the Mailing FIFO)
!LN	The LJ3 sends in the protocol the length
!EM	Switch on the echo mode

Details to commands

!RC

Reset all counters to its initial value which are defined in a job as counter reset source. This function has no effect on the product counter (for settings of product counter please see =CC)

Example: ^0!RC 0x0D

Look at the command =RC.

!LF

This instruction is necessary if fonts have been changed or added and which should be used by the current job.

Attention! This instruction aborts the print shortly.

!US

This instruction prevents user - inputs for 10 seconds (information window on the operator interface opens).

If the interface should be blocked for the operator for the duration of a remote connection, the remote application has to repeat this instruction within 10 seconds continuously or a communication should be proceeded within 10 seconds (inquire status, read files etc.) that means the user can operate the device again after activation of the suspend mode and a communication break of >10 seconds. This function should provide an operation even in case of a connection abort. By sending the !UR command you can leave this mode and the operation will be enabled immediately.

!EQ

Provides the acknowledgement of arising errors at the machine.

Errors have to be acknowledged (especially in the User Suspend mode) because the machine will turn off after a certain period of time due to safety reasons. If and which errors are occurred can be determined with ?RS (see there).

!LN

Since V60.0.00.56

The LJ3 sends from this moment in his protocol the length of data. This will be done until the LJ3 will be switched off or he gets a factory reset command.

The protocol is not changed for the receiving of the Leibinger-Jet3. Here the length will not be send.

The length is always a decimal number with 5 digits and comes directly after the address byte and before the command group character.

This length defines the number of bytes which are coming from now on including the terminating carriage return.

Example:

Host sends:	LJ3 answers:
^0?RS <CR>	^0=RS2 6 0 0 0 <CR>
^0!LN <CR>	
^0?RS <CR>	^000013=RS2 6 0 0 0 <CR>

!EM

Since V60.0.01.4D

Switch on the cho mode. This remains switched on until the LJ3 will be switched off or he gets a factory reset command.

Function: The LJ3 sends all action- and parametercommands directly after he received it back to the host. The LJ3 does this only, when he can process this command. These are the commands with the ! and = in the command group.

Inquiries

With an inquiry you can demand parameters from the opposite side.
 The opposite side responds on an inquiry with the according =XX parameter command.

Overview of inquiries (Host → Machine)

Command	Description
?EX	Report Extern-States
?SH	Report Service History
?IC	Report Input Configuration
?HS	Report Hydraulic States
?OC	Report Output Configuration
?MC	Report Main Configuration
?CC	Report Current Counter
?SC	Report Service Configuration
?VS	Report Version
?JL	Query name of loaded job
?JL	Query name of loaded camera job
?EL	Query ErrorList
?EV	Query EventList
?JB	Inquiry for the current print job. This inquiry is sent from host to the machine, hereupon the machine sends the job (in ^0*BEGINSCRIPT.... blocks (see above)
?SM	GetMailStatus
?DL	Query date/time
?RS	Query status of data
?GW	Query Wysiwyg
?ET	Extern Text

Parameter Transfer

With the help of parameter-transfer commands you can send current settings and status-information from the machine to a host or new settings can be transferred from host to the machine. The single parameters are separated with <TAB>-characters. In case a parameter is empty, the current value will not be changed.

The following parameter-instructions are implemented:

=AC All Counter (Machine → Host)

Par1: Number of counter used in the current job

A job can have up to 32 counters.

For each counter there are 3 paramters:

Counter n:

Par $3*(n-1)+2$: Done repetitions

Par $3*(n-1)+3$: Countervalue High

Par $3*(n-1)+4$: Countervalue Low

Parameter Details:

Done repetitions

This is the value how often this counter value was printed. When the counter parameter "repetitions" is 0, this value is always 1.

Counter value High und Low

Because a counter can have up to 10 figures, there are 2 32 bit values necessary.

=CC Current Counter (Machine ↔ Host)

Par1: Product Counter

Par2: Stop after X Products

Par3: Total print counter

Parameter Details:

Product Counter

Current status of product counter.

Stop after X Products

With reaching this product counter status the print is stopped automatically by the LJ3. (Exception 0 = function deactivated)

Total print counter

The total print counter can only be read (read only) and not be changed.

His value is the number of prints which this Jet3 has done.

=CM Config Mail (Machine ← Host)

Par1: StopAtNo

After sending this parameter the LJ3 stops the print automatically if this mail record number has been printed.

Exception is 0 (default). No print stop is carried out for 0.

=CO Change Output (Machine ← Host)

Par1: No. of the output which should be changed. Range of value: 0...8

Par2: Wished state of the output. Range of value: 0 or 1

The Jet3 has the outputs 1...8.

Special case: Output No. 0:

^0=CO0 1: Command Initialising

^0=CO0 0: Command End. The Jet3 now controls the outputs itself again.

Before with this command could be worked, at first one time the command "Initialising" should be sent. Only in this case the settings made in I/O Settings for these outputs believe conserved for which no CO command will be sendet.

To avoid conflicts, the outputs which will be changed with CO, should not be used in the I/O Settings.

Attention: The output „Warning“ is not available during working with CO.

Example:

^0=CO0 1 Initialising.

^0=CO3 1 Set output 3 to 1. Output „Warning“ is not available.

...

^0=CO0 0 End. Output 3 and „Warning“ are in the original state again.

=DL Date/Time (Machine ↔ Host)

Par1: Year

Par2: Month

Par3: Day

Par4: Weekday (0-6, 0 = Sunday)

Par5: Hour

Par6: Minute

Par7: Second

=

=EL ErrorList (Machine → Host)

The error list is transmitted in blocks and every block has max. 32 entries. The

ErrorList contains:

Par1: Number of first entry (1...)

Par2: Total number of entries

Par3: Number of entries in this block

Par4: Entry1

Par5: Entry2

.....

Every entry is transmitted in the form `yy-mm-dd hh:mm:ss/errornumber`

=ET ExternText (Machine ← Host)

Par1: String

String = External text with up to 256 characters and <CR> for final identification

Reaction time of the LJ3: 20...40msec

=EV EventList (Machine → Host)

The event list is transmitted in blocks and every block has max. 16 entries. The

EventList contains:

Par1: Number of the first entry (1...)

Par2: Total number of entries

Par3: Number entries in this block

Par4: Entry1

Par5:

Every error entry is transmitted in the form

`yy-mm-dd hh:mm:ss/cp/ss/st/mr/pr/dp/ft/ms`

with

cp Compressor-Pulse

ss Sips of solvent

st Sips of ink

mr MixRatio

pr Pressure

dp Diaphragma-Position

ft FillTime

ms MachineState

=EX Extern States (Machine → Host)

Par1: Actual Speed

Par2: Inputs (Integer number from 0 – 2¹⁷)

Par3: MaxSpeed

Par4: Direction

Par6: PrintGo

Par7: Reserved

Par8: Reserved

Par9: Reserved

Parameter details:***Actual Speed:***

Information of the current printing speed in m/min

Inputs:

Bit 0 = X4.9

Bit 1 = X4.22

Bit 2 = X4.10

Bit 3 = X4.23

Bit 4 = X4.11

Bit 5 = X4.24

Bit 6 = X4.2 (Counter Increment)

Bit 7 = X4.1 (Counter Reset)

Bit 8 = X4.3 (Job select 0)

Bit 9 = X4.16 (Job select 1)

Bit 10 = X4.4 (Job select 2)

Bit 11 = X4.17 (Job select 3)

Bit 12 = X4.5 (Job select 4)

Bit 13 = X4.18 (Job select 5)

Bit 14 = X4.6 (Job select 6)

Bit 15 = X4.19 (Job select 7)

Bit 16 = X4.7 (Job select 8)

Bit 17 = X4.20 (Job select 9)

Max Speed:

Maximum possible printing speed for current setting in m/min.

Direction:

Encoder direction:

0 = Backward

1 = Forward

PrintGO

0 = PrintGo Sensor Signal is Low

1 = PrintGo Sensor Signal is High

Reserved:

Always 0

=FC CRC32 failed (Machine → Host)

Par1: CRC32 Checksum which has been calculated by the machine and which did not correspond with CRC32 transmitted with =NR.

=FL Fix Blocklength (Machine ← Host)

since V60.0.00.56

Par1: Blocklength in bytes

With this command the LJ3 will be adjusted to send with a fix block length. All data, which the LJ3 sends, will be filled up to this block length with blanks. The block length can be adjusted in the range between 1 and 2048. It will be active until the next power on of the Ljet3.

=GW Wysiyg Data (Machine → Host)

The response happens block by block with max. 60 characters per line. For final identification the sender (machine) transfers a ^0!EN.

Data layout:

60 characters per line, nibble coded that means 30 bytes per line

Coding High Nibble (Bit 7-4) 0-15 = A-P

Coding LowNibble (Bit 3-0) 0-15 = A-P

Byte 1-16 Header information:

Byte 1-4 = Overall width of bitmap

Byte 5-8 = Height of bitmap in pixel

Byte 9-12 = Job ID of this bitmap (for ext. Jobselect the applied ext. number)

Byte 13-16 = Number of actual delivered strokes

Byte 17 till end: Stroke data

=HS Hydraulic State (Machine → Host)

Par1: ActFalltime

Par2: ActPress

Par3: ActSuctpump

Par4: Diaphragm Position

Par5: Mix-Value

Par6: State

Parameter details:**Act Falltime:**

Current fall time of the viscometer in ms.

ActPress:

Current pressure in mBar

ActSuctpump

Current suction pump value in %

Diaphragm position

Current diaphragm position of the pressure tank.

(Dimensionless, normal operating positions between 700 and 1400)

Mix-Value

Current solvent mix-value in ink to solvent.

0 = no solvent admixture

State:

Not documented!

=IC Input Configuration (Machine → Host)

Par1: Input1 (assigned X4.9 function)

Par2: Input2 (assigned X4.22 function)

Par3: Input3 (assigned X4.10 function)

Par4: Input4 (assigned X4.23 function)

Par5: Input5 (assigned X4.11 function)

Par6: Input6 (assigned X4.24 function)

Parameter details:

Decoding of numbers to the assigned function:

0 = Disabled

1 = Nozzle Open

2 = Nozzle Close

3 = PrintStart

4 = PrintStop

5 = Printer Shutdown

6 = Horizontal Mirror

7 = Cutting Synchronisation

=JL JobLoad (Machine ↔ Host)

Machine->Host: The Ljet sends the current Jobname with path.

Host->Machine: The Ljet loads the job with the name in Par1.

Reaction time of the LJ3: 50...150msec

Par1: Jobname (with path information)

Example: ^0=JLFFSDISK\JOBS\TESTPRINT.JOB 0x0D

=JC JobLoad for Camera Job(Machine ↔ Host)

Machine->Host: The Ljet sends the current Jobname with path.

Host->Machine: The Ljet loads the job with the name in Par1.

Par1: Jobname (with path information)

Example: ^0=JCFFSDISK\JOBS\TESTPRINT.JOB 0x0D

=MC Main Configuration (Machine → Host)

Par1: DirectPrint

Par2: Encoder Setting

Par3: Reserved (0)

Par4: Reserved (0)

Par5: Reserved (0)

Par6: Set Fall Time

Par7: Set Pressure

Par8: Reserved (0)

Par9: Set Visco Correction

Par10: System Frequency

Par11: Inknumber

Par12: Interval Time on

Par13: Interval Time off

Parameter Details:***DirectPrint:***

0 = Direct Print function deactivated / 1 = Direct Print function activated

Encoder Setting:

0 = RS422

1 = TTL (5V level)

2 = HTL (24V level)

Set Fall Time

Current set target fall time of the viscometer in ms

SetPressure

Current set system pressure in mBar

SetViscoCorrection

Current set viscometer correction factor

SystemFrequency

Dot frequency in kHz

InkNumber

Ink-Number which is filled in the printer.

Interval Time ON / Off

Interval times in the following format:

Hour*100+minutes

Hours 0-23 and minutes 0-59

=MR Mail Record (Machine ← Host)

Par1: String

The string is assembled as following (e.g. 3 Field Record):

No + tab + Field1 + tab + Field2 + tab + Field3+ ... + tab + {[h][ox][rn]} 0x0D

Example: ^0=MR17 Miss Jane Miller {ho2r3}0x0D

When in the job of the Leibinger-Jet3 is a graphic which uses mailing, the corresponding text field should contain a graphic. The format of the graphic is described in the document "Manual_LJ3_Interfaceprotocol.pdf" in the chapter OBJ Parameter list.

Example: ^0=MR0 111 (8 3 FF81FF) 0x0D

Definitions:

No. is a displayed ASCII consecutive 32 Bit No. without any prepend zeros and has to exist coercively !

Field should not contain ^ or has to be replaced by \^ !

A record can contain up to 255 fields.

Since V60.0.1.22 there are the **Property-Parameter** in this braces:: {}. These are after a tab directly before the 0x0D.

The brackets [] are not used. They have the meaning that these parameters inside of it are optional. Each of these 3 parameters may be used only one time.

h = horizontal mirroring of this record
 o0 = Orientation 0°
 o1 = Orientation 90°
 o2 = Orientation 180°
 o3 = Orientation 270°
 rn = n repetitions of this record

The parameter repetition may only be used when the 1. Parameter, that is the Number No., is always 0.

Examples:

^0=MR0 X13.1 {hvr3}0x0D

$\wedge 0 = \text{MR0}$ X14.2 {r21}0x0D

=NR Next CRC32 (Machine $\leftarrow$ Host)

Par1: CRC32 Checksum of the proximate communication string.

=OC Output Configuration (Machine $\rightarrow$ Host)

Par1: Output1 (signal assigned to X3.1)
 Par2: Output2 (signal assigned to X3.2)
 Par3: Output3 (signal assigned to X3.3)
 Par4: Output4 (signal assigned to X3.4)
 Par5: Output5 (signal assigned to X3.9)
 Par6: Output6 (signal assigned to X3.10)
 Par7: Output7 (signal assigned to X3.11)
 Par8: Output8 (signal assigned to X3.12)

Parameter Details:

Decoding of No. to the assigned signal

0 = Disabled
 1 = Refill Warning
 2 = Error/Print Stop
 3 = Print Readiness
 4 = Print active
 5 = Product counter reached Value
 6 = PrintGo
 7 = Encoder Signal (left/right)
 8 = Power fail out
 9 = Counter reached end value

=PR Print Parameter (Machine $\leftarrow$ Host)

since V60.0.01.04D

Par1: Print parameter type, which could be changed

Par2: New value for the parameter

With this command a single parameter in the Jet3 can be changed.

Par 1:	Meaning of Par1:	Unit of Par 2:	Time of the result:
1	Print Go delay	Mikrometer. 1mm=1000µm	With the 2. PrintGo
2	Print Go distance	Mikrometer. 1mm=1000µm	With the 2. PrintGo
3	Strokedistance	Mikrometer. 1mm=1000µm	With the next PG
4	Printing hight	30 ... 100	With the next PG

5	Speed of the internal encoder pulses	In dm/min *)	With the next PG
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Example:

^0=PR2 31000 Sets the PrintGo distance to 31mm (31000µm)

*) from V60.0.1.17. before: m/min

=RC Reset Counter (Machine ← Host)

since V60.0.01.02

Par1: ID-Number of the counter

Par2: Start value

Reset counter means that the counter will be set to the start value.

Par1 contains the ID-Nr. of the counter, which have to be resetted. If Par1=0, all counters are resetted.

With Par2 you can send the startvalue. If this value is -1 the startvalue will be the startvalue of the job. Otherwise the counter will be resetted to this parameter. If no Par2 is given, Par2 = -1 (Jobsetting)

=RS Machine State Information (Machine → Host)

Par1: Nozzle state

Par2: State of Machine

Par3: Error

Par4: Headcover

Par5: ActSpeed

Definitions:

Nozzle state:

0 = invalid (Standby or initialization)

1 = opens

2 = is open

3 = closes

4 = is closed

5 = inter, in between

State of Machine

1 = Standby (The other values e.g. nozzle state are invalid !)

2 = Initialization phase (incl. Bleeding)

3 = Interval or Service Panel

4 = Ready for Action (ready for operation, instructions as e.g. nozzle open can be carried out)

5 = Ready for Printstart (ready for print start, that means nozzle is open and oscillator voltage is set)

6 = Printing

Error:

0 = no error

xxx = Error number

Besides the error code further information regarding the error source and malfunction are defined in the error number:

Bit No.	Meaning	Values
26/25	Error source	00 = FEP CPU. (User interface) 01 = RIP CPU (Print processing + hydraulic) 10 = SDC CPU (Dots + Print head control)
27	Malfunction Shutdown	0 = Device turns off after 30min. if no confirmation happens. 1 = Device turns not off
29/28	Malfunction Signal tone transmitter	00 = Permanent signal tone 01 = one-time signal tone 10 = no signal tone
31/30	Malfunction Display	00 = Display error window 01 = Display warning window 10 = Display message window

After masket reading of the bits 25-31 one error code remains which specifies the occurred error. Error texts which should be displayed have to be seen in the error*.txt data files in the folder FFSDISK\LJ3. (Therein you will find a language dependent decoding of the error numbers to the error text)

Headcover

0 = Headcover closed

1 = Headcover open

ActSpeed

In m/min

=SC Service Configuration (Machine → Host)

Par1: Service Hour Interval

Par2: Service Year Interval

Parameter Details:

ServiceHourInterval:

Operating hours between two service cycles

ServiceYearInterval:

Days between two service cycles

=SH Service History (Machine → Host)

Par1: Operating Hours

Par2: Last Service Hours

Par3: Last Service Date

Parameter details:**Layout Last Service Date (32 Bit):**

Bit 31-26: Year-2000

Bit 25-22: Month (1-12)

Bit 21-17: Day of month(1-31)

Bit 16-12: Hour (0-23)

Bit 11-6: Minute (0-59)

Bit 5-0: Second (0-59)

=SM Mail Status (Machine → Host)

Par1: Fifo Depth

Par2: Fifo Entries

Par3: LastStartedPrintNo

Par4: StopAtNo

Par5: LastStartedPrintNoWasFinished

Description of parameters:

FifoDepth = Available batch depth of the Mailing Fifos in the LJ3. (Normally 256 records)

FifoEntries = Number of mail records which are available in the mailing Fifo

LastStartedPrintNo = No.of the mail record which's print has already started
(0 =if no one is available)

LastStartedPrintNoWasFinished

= Flag (If = 0, the last print was started, but not finished,
if >0, the last print was finished, but the next one for sure not
started)

StopAtNr = Record number where a print stop is generated
(=Value which has been set before with ^0=CM)

To be sure that this parameters are actual it is better when the host sends the command ^0?SM twice to the Leibinger-Jet3. The second answer of the Leibinger-Jet3 is always actual.

Please recognize that the value for FifoEntries always is reduced by one as the number of records you actually have sent. Reason: The record which is in printing is already removed.

=VS Version (Machine → Host)

Par1: FPGA-Version
Par2: RIP-Software Version
Par3: SDC-Software Version
Par4: FEP-Software Version
Par5: KernelVersion
Par6: SystemType
Par7: LanguagePack
Par8: FontPack
Par9: LJVersion of firmware
Par10: Type
Par11: Serialnumber

Parameter details:

All parameters are tab separated strings

File Transfer

In the FileTransfer-area data and directories are exchanged. The following instructions are supported:

\$RD Request-Directory (Machine ← Host)

Sends a Dirinfo inquiry to the machine.
The machine responds with a DI-info.

Par1: Directory string

In the directory string also the common substitution characters are valid (Joker *)
Some examples:

Existence of the job check directories:

Inquiry: ^0\$RDFFSDISK\Jobs

Response: ^0\$DI1 01 !Jobs

(= Directory exists)

Inquiry: ^0\$RDFFSDISK\Jobis

Response: ^0\$DI1 00

(= this directory or this data file does not exist)

Readout all data filenames of a directory

Inquiry: ^0\$RDFFSDISK\Jobs*

Response: ^0\$DI1 06 COUNTER.JOB DATE.JOB DM.JOB Testprint.job
LINE.job LJ24 Test.job

Display all ISO Fonts in the fonts directory

Inquiry: ^0\$RDFFSDISK\Fonts\ISO*.lft

Response: ^0\$DI1 09 ISO1_5x5.lft ISO1_7x5.lft ISO1_9x7.lft ISO1_12x8.lft
ISO1_14x10.lft ISO1_16x10.lft ISO1_24x18.lft ISO1_32x20.lft
ISO1_5x7flue.lft

Standard directories in the LJ3

Data carrier	Memory type	Path
Internal Flash Disk	Jobs	FFSDISK\Jobs
	Fonts	FFSDISK\Fonts
	Graphics	FFSDISK\Graphics
SD Card	Jobs	Storage Card\LJ3\Jobs
	Fonts	Storage Card\LJ3\Fonts
	Graphics	Storage Card\LJ3\Graphics
USB Stick	Jobs	Hard disk\LJ3\Jobs
	Fonts	Hard disk\LJ3\Fonts
	Graphics	Hard disk\LJ3\Graphics

\$DI Directory-Info (Machine → Host)

The directory-info is transmitted in blocks and every block has max. 32 entries. The DirInfo contains

Par1: 0= further blocks follow, 1=last block

Par2: Number of entries in this block

Par3: Entry1

Par4:

Every single entry has a data filename.

\$RF Request File (Host → Machine)

Sends a FileTransfer inquiry to the machine.

Par1: Data file with directory information

To copy a job of the internal Flashdisk to the host, you have to send:

^0\$RFFFSDISK\Jobs\MyJob1.job<CR>

The machine responds with FH and FT-instructions.

\$FH File Header (Host ↔ Machine)

File-transfers are implemented from host to machine as well as from machine to host. Every FileTransfer has to be started with a Header. It contains:

Par1: FileName with path. As path is always transferred the path on the machine side.

Par2: Number of the following FT-blocks. In case a 0 is transferred, the required data file cannot be opened.

\$FT File Transfer (Host ↔ Machine)

With this instruction you can transfer a File Transfer block.

Par1: Blockno

Par2: Blockcheck (=Checksum modulo 256 of all data bytes of this block)

Par3: Data.

The data are transmitted as nibble-string with 4 bit per character. The first nibble has the bits 7-4 of the first byte, the second the bits 3-0, the third nibble the bits 7-4 of the second byte etc.. The nibbles are coded as values from A...P (A=0, P=15).

After the receiver has analyzed the package, a file Acknowledge (FA) has to be returned.

\$FA File Acknowledge

Confirmation of received file package

Par1: Blockno.

Par2: 0=transmission error, block has to be repeated

1=understood, Block okay

2=Cancel complete File-transfer

Transmissions with CRC32

To guarantee a secure data transfer also with a RS232 transmission and interfering field environment, the protocol offers a CRC32 procedure.

This support does not apply to the file transfers (for this separated block check numbers are used. See \$FT File Transfer)

If a transmission with CRC32 should be carried out is always initiated by the host prior to every transmission.

For the procedure of a CRC32 transmission the following commands and parameter transfers are important:

ParameterTransfer: NextCRC32 (Implementation Host<->LJ3)

^0=NR Par1 = CRC32 Checksum

ParameterTransfer: CRCFailed (Implementation Host<-LJ3)

^0=FC Par1 = CRC32 Checksum

Command:: CRCok (Implementation Host<-LJ3)

^0!OK

Description of operation:

If you would like to have a secured transmission, the host calculates previous to every transmission the CRC32 checksum of the next transmission string and transfers it with ^0=NC to the LJ3.

It memorizes this CRC32 checksum, calculates and compares the CRC32 checksum of the transfered string for the next incoming communication string.

If both are unequally a response with the checksum which has been calculated in the LJ3 will be sent to the host ^0=FC.

The host is then able e.g. to carry out a transmission repetition.

If the checksum is okay the LJ3 responds (prior to the transmission of possible data) with ^0!OK.

If an inquiry of data (^0?XX) with CRC32 has been sent to the LJ3 a CRC32 checksum is transfered from the LJ3 to the host prior to every response line.

Example of a (Hyperterminal) sequence with CRC32:

Query of the current used jobs (H = Host, L = LJ3):

<i>^0=NR3957421711</i>	➔ CRC32 of the following ^0?JL inquiry (H->L)
<i>^0?JL</i>	➔ inquiry itself (H->L)
<i>^0!OK</i>	➔ the confirmation of the LJ3 the CRC OK (H<-L)
<i>^0=NR3560773416</i>	➔ CRC32 of the pending response (H<-L)
<i>^0=JL\FFSDISK\JOBS\Testprint.job</i>	➔ response itself (H<-L)

Mailing (Database) operation

The commands and parameter transfers which are relevant for the mailing operation are:

^0=MR String

^0=CM Par1

^0?SM and accordant response from LJ3 ^0=SM

^0!FF

Info regarding performance/Procedure:

- For a print stop all FIFO contents and parameters are deleted (StopAtLfd = 0) Texts which are transmitted by the protocol can be changed in length. That means the „Wildcars“, which are set in the editor are only used for the editor after rebooting of the job or by sending a “Zero” string.
If a fixed text length is required this has to be done in the application which sends the mailing text. (is also valid for “Extern Text”)
- In the LJ3 all mail messages are checked for ascending numbering during readout of the FIFO and in case this regulation can be not observed a „Warning“ message is generated. Exceptions are mail records with 0.
- If the StopAtLfd value is >0, the print will be automatically stopped by reaching this number.
- The LJ3 checks the numbering with each PrintGo(PG). With RecordNo 0 no check will be done and when the mailing buffer gets empty, with each PG the last received record will be printed once more. When the record numbers are numbered, this numbering will be checked. When this is not correct, a stop print and an error will be caused. When there is a numbering, the mailing buffer may not get empty, otherwise an error and a Stop print will be caused. Around 3 Milliseconds before a PG is coming, the mailing buffer must have at least one record. Is the speed 180 m/min or more, the mailing buffer must have at least one record 3 Milliseconds before the last stroke is printed.

Due to the above described performance the following is applied for the consecutive record number:

1. It has to be always available
2. It has to be 0 or numbered in ascending order and serially (No = No+1)
3. The first number of a new number sequence has to be always 0.

Example: 0 0 0 0 500 501 502 503 0 100 101 102 ...

Hint:

If Extern Text or Mailing operation is carried out with Unicode Fonts (defined in the accordant job) the text data has to be transferred hexadecimal coded!

e.g. "Hello World" has to be transferred as following
00480065006C006C006F00200057006F0072006C0064

History

V1.3	Command !LN added
V1.4	Command =FL added
V1.5	Command =RC added
V1.6	Command !EM added Command =PR added
V1.7	Command ?CC: Total print counter added
V1.7.1	Command =AC added
V1.7.2	Parameter 9 in command =OC added
V1.7.3	New: Command ?ET
V1.7.4	Command PR: Speed at internal encoder pulses is now dm/min
V1.7.5	Command =MR: Grafik Mailing added (since V60.0.1.22)
V1.7.6	Command =MR: Property-Parameter added (since V60.0.1.22)
V1.7.7	Command =CO implemented (since V60.0.25.0)